SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Assessment Tool Weightage: 40%

Total Marks: 30

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Code of Conduct
I K.Sumanth Kumar Reddy (192425222) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: K.Sumanth

Assessment Tasks
Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1) Unified Big Data Architecture for Multinational Retail

A multinational retail organization may have customer data stored separetely in websites, mobile apps, physical stores and social media. This fragmentation makes it difficult to obtain a complete view of customer behaviour.

A suitable architecture is:

Data Sources → Data Ingestion → Data lake → Processing → Unified customer profile → Analytics/ML → Recommendation

- Data sources: online purchases, POS transactions, mobile apps, websites, loyalty cards and records.
- Data ingestion: Apache Kafka can collect real-time transaction and clickstream data.
- Data storage: Hadoop HDFS or cloud object storage can store large volume of structured data.
- Processing: Apache spark can perform batch and processing.
- Data integration: Customer ID's, email ID's, and loyalty ID's can be matched to create customer profile.
- Analytics: machine learning can identify customer preferences, purchase patterns and customer segments.

This architecture provides scalability, a real-time processing and a 360-degree view of customers, enabling personalized recommendations.

2) Distributed computing for scientific Research

Scientific reasearch generates huge datasets from sattilites, medical experiments, simulations, gnemonics and particle physics. Processing such data using a centralized system can be slow and expensive.

Centralized System:

- one main computer processes the entire dataset.
- Limited processing capacity
- Higher processing time
- Single point of failure
- Difficult to scale

Distributed system:

- Data and computation are distributed across multiple nodes.
- Tasks execute in Parallel
- Processing time is reduced
- Additional time can be added easily
- Failure of one node does not necessarily stop the complete system.

Technologies such as Hadoop MapReduce and ApacheSpark support distributed processing.

Therefore, Parallel Processing improves speed, Scalability making distributed computing more suitable for large-scale research

3) Big Data Fraud Detection Framework

A digital payment platform recieves millions of transactions from different locations and devices. Traditional fraud detection may not identify fraudulent transactions quickly enough.

A suitable framework is:

Transactions → Kafka → Stream Processing → Feature Engineering → ML Model → Fraud score → Alert/Block

- Data sources: Transaction amount, location, device time, customer history and previous fraud records.
- Kafka: collects payment transactions in real time
- Spark Streaming: Processes transaction continuously
- Feature engineering: calculates unusal transaction, location changes, transaction frequency.
- Fraud score: Each transaction recieves a probability.
- Action: High-risk transactions can be blocked or sent for manual verification

Streaming processing enables fraud to be detected within seconds rather than after batch processing.

Thus, combining real-time analytics with machine learning improves fraud detection accuracy.

4) # Multimedia Data in Big Data

multimedia data includes videos, audio files and images. These datasets create several challenges because they are large, unstructured and expensive to process.

Major Challenges:

- very large file sizes
- unstructured data formats
- High storage requirement
- complex processing and analysis.

Suitable Technologies

Storage:
- Hadoop HDFS
- Cloud object storage such as Amazon S3
- Distributed file systems
- NOSQL databases for metadata

Processing:
- Apache spark
- Hadoop MapReduce
- GPU-based Processing

Analysis:
- computer vision for images and videos.
- speech recognition for audio.
- Deep learning for classification and object detection.

A practical solution is to store multimedia file in distributed object storage while storing medadata.

5) Disaster Recovery and Business Continuity

A cloud based big data platform must continue even when hardware, software, network or cloud-region failures occur.

A suitable strategy include:

Primary system → Replication → Backup/secondary system → Automated failover

Key components

1. Redundancy: Maintain multiple servers and systems.
2. Data Replication: Replicate important data sets across multiple nodes or regions.
3. Regular backups: Create scheduled backups and maintain recovery copies.
4. Disaster recovery site: Maintain a secondary environment.
5. Automated failover: Automatically redirect applications to healthy systems.
6. Monitoring: continuously monitor system health.
7. Recovery testing: Regularly test restoration and failover procedures.

Replication prevents data loss, while redundancy prevents a single hardware failure from stopping the system. Automated failover reduces downtime.

6) Data Quality and Data Governance

Poor-quality data can seriously affect the reliability of big data analytics.

Common Data Quality Problems

- Inconsistency: Different systems use different formats or values.
- Duplication: The same customer or transaction appears multiple times.
- Missing values: Important fields may be unavailable
- Incorrect values: Data may contain errors.

Data Governance Strategies:

- Establish data quality strandards.
- Use master data management
- Remove duplicate records
- Validate data during ingestion
- Handle missing value using appropriate techniques
- Maintain data dictionaries and metadata
- Define data ownership and responsibilities
- Continuously monitor data quality
- Implement access control and auditing

Good data governance, ensures that analytical results are accurate, consistent, reliable and trustworthy.

7) How does Apache NiFi help?

→ NiFi helps move and integrate data between different systems.

→ It can collect, filter, transform and route data.

→ It also provides data tracking and monitoring.

8) Failure points in Hadoop:

→ common failures include Data nodes, disk, network and tash failures.

→ Hadoop handles failures using replication and task recovery

→ monitoring helps identity problems quickly

Limitations:

- Data security and privacy risks
- Dependence on cloud providers
- Network dependency
- Unexpected cloud costs
- Vendor lock-in
- compliance challenges

Therefore, cloud computing provides excellent scalability and flexibility, but organizations must carefully manage security, costs and compliance.

9) Real-Time Transportation and Logistics Analytics

Transportation companies generate data from GPS devices, vehicles, traffic systems, weather services.

A suitable architecture is

GPS/vehicle sensors → Kafka → Streaming processing → Analytics/ML → Dash board/Alerts

Applications

Route optimization:
Real-time traffic and GPS data can identify route.

Fuel Efficiency:
vehicle speed, engine performance, distance and driving Patterns

Predictive maintainance:
sensors data can identify possible vehicle failures.

As a result, organizations can reduce travel time, fuel consumption and operational costs while improving decisions.

10) Ethical and legal challenges of Big Data

Big data provides valuable insights but creates significant ethical and legal concerns.

Ethical challenges:

- Loss of user Privacy
- Excessive data collection
- Lack of informed consent

legal challenges:

organizations most comply with applicable data protection and privacy regulations. They must properly manage personal information, data retention and security

Recommended Policies:

1. Collect only necessary Data
2. obtain informed user consent
3. Encrypt sensitive information
4. Apply strong access controls

Responsible data governance improves regulatory compliance, privacy, fairness and customer trust.